Rewind & Amplify

Predicting how hidden defects change quantum measurement patterns

A Qiskit model tested on IBM hardware.
Exploratory extension: laser diagnostics.

Qubits / Florida Gulf Coast University

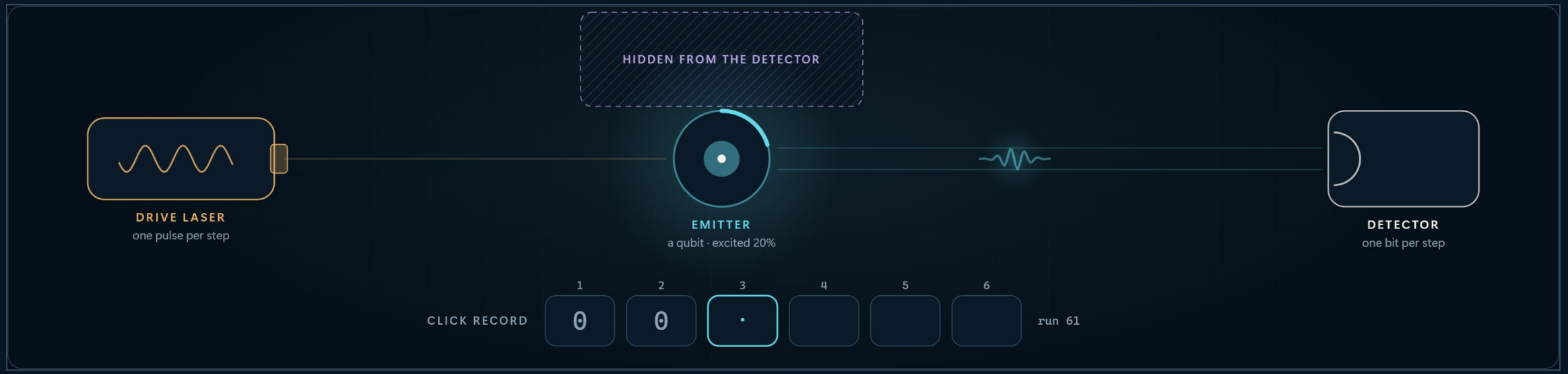
Florida Qiskit Hackathon finals / FAU / 14 November 2026



THE PROBLEM

Something is hidden inside the signal.

A detector records one bit per time step: click, or no click. Out of sight, a defect can trade energy with the source and quietly change the pattern.



Quantum-chip teams

Unseen defects drain their qubits between calibrations.

Photon-source engineers

Two sources can look equally bright and behave differently.

Anyone reading a detector

A detector fault can look exactly like a defect.

Guess the cause wrong, and you fix the wrong thing.

Picture of our model, captured from the live Lab. On IBM hardware, superconducting qubits play these roles; no optical photons were detected.

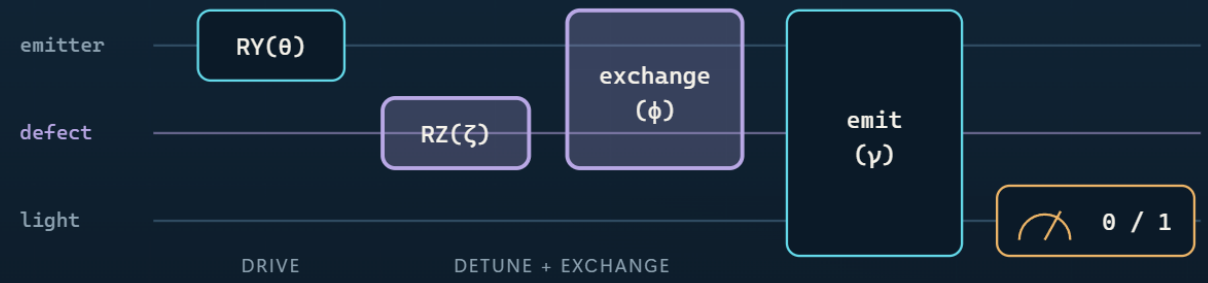
OUR MODEL

A recurrent machine whose memory is quantum.

We call it a QRSM: a quantum recurrent state machine. Every time step repeats the same four gates.



ONE STEP, GATE BY GATE



The animation above is not a cartoon. It runs these gates, and a test holds it to the Qiskit code.

2 qubits

of memory: the emitter and a hidden defect. Never measured.

1 bit

comes out per step: click, or no click.

Each bit

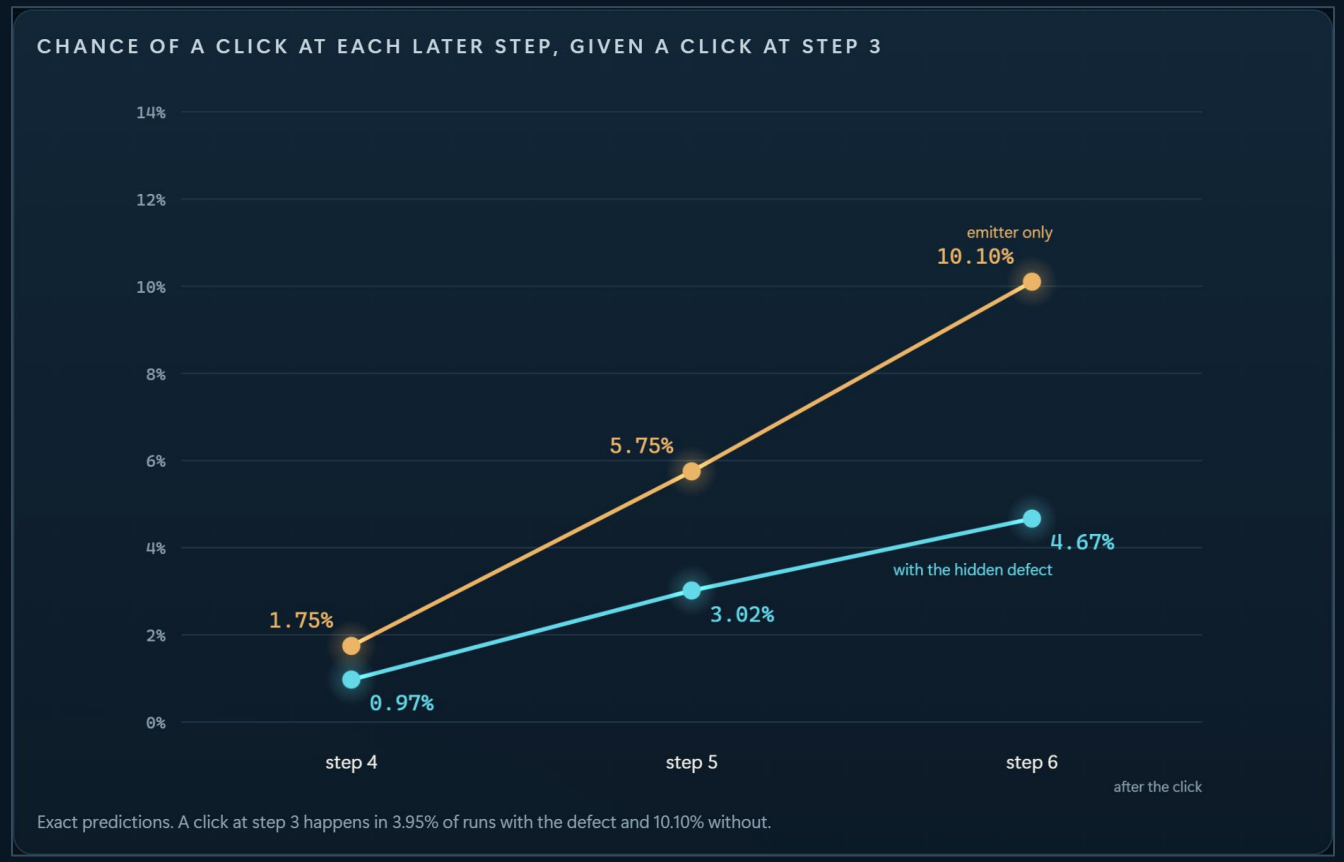
updates the hidden state, so the model predicts what comes next.

Same gates as qrsm_grover/model.py. The picture is a live simulation of that circuit, tested against Qiskit to 10^{-12}.

THE QUESTION

One click. Two futures.

Given a click at step 3, what happens next? With no defect, a click empties the source and the machine starts over. A hidden defect remembers.



EMITTER ONLY

10.10%

chance of a click at step 6

WITH THE HIDDEN DEFECT

4.67%

chance of a click at step 6

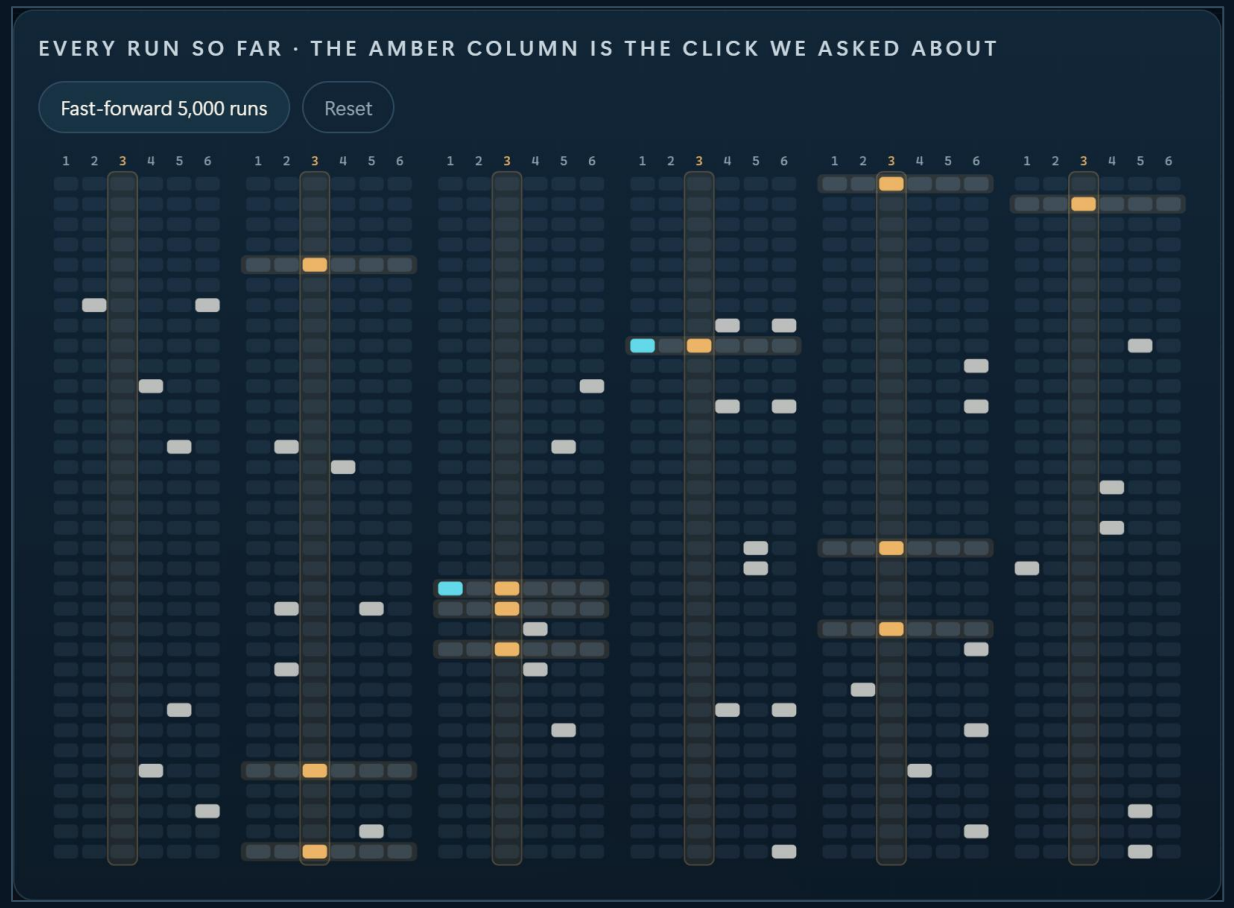
The pattern after a rare click is a fingerprint of hidden memory.

Exact predictions of two models we programmed, recomputed live by the Lab and by reproduce.py. Not a diagnosis of a real defect.

THE COST

The useful runs are the rare ones.

To study what follows a rare click the ordinary way, you run the machine again and again and throw away every run that missed.



3.95%

of runs contain the click we asked about

25 runs

per useful sample. The other 24 are thrown away.

27,495,891 runs

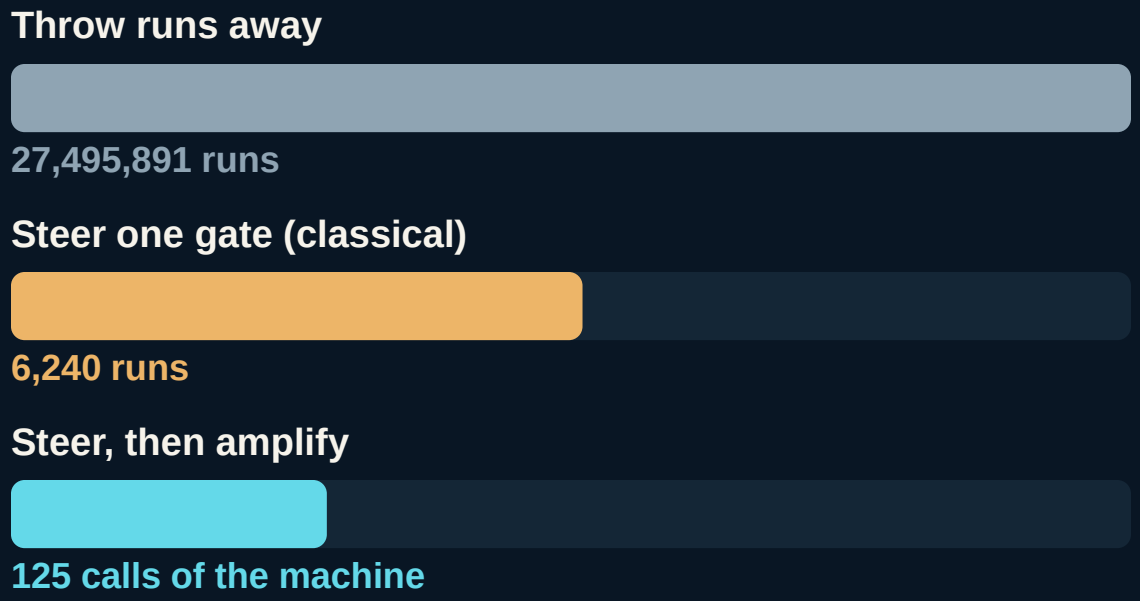
to match one real 64-step record this way

Rejection sampling costs 1/p runs per kept sample. The 64-step figure is an exact calculation for a real record from our first project.

WHY QUANTUM MIGHT HELP

Grover makes rare answers common. Two catches.

COST OF ONE MATCHING SAMPLE · 64-STEP RECORD · LOG SCALE



Rarity costs 1/p the ordinary way, and about 1/√p with amplification.

1 **It must run the machine backwards.**

Our model resets a qubit every step, and a reset cannot be undone.

2 **It is deep.**

On today's hardware every extra gate is a chance for noise.

No speedup today. At this size an exact classical sampler answers in a fraction of a millisecond. Its cost grew about 6× per added hidden qubit in our tests: that is where a quantum method could matter.

Amplitude amplification is prior work (Brassard et al. 2002; Low, Yoder and Chuang 2014). Bars are exact calculations for one 64-step record, not hardware runs.

OUR METHOD

You can’t amplify what you can’t rewind.

1 Purify

A fresh qubit replaces every reset. The whole run becomes one reversible circuit.

2 Steer

Tune one gate so the rare click is less rare. The answer is unchanged.

3 Amplify locally

One round. The reflection checks 3 qubits, not all 5: about half the gates.

4 Verify

The click is measured directly, so a failed shot shows itself and is dropped.



IDEAL SIMULATOR · ONE ROUND

3.95% → 96.1%

chance of the click we asked about

0.000

error in the answer with the full reflection

0.035

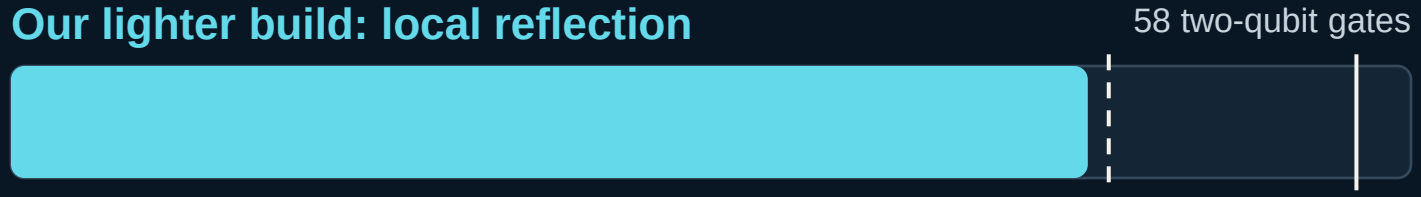
with the lighter local reflection: a small, known bias

What is ours: the mapping onto a hidden-memory model, and the noise-aware build. Qiskit circuits, Aer verification, transpiled to IBM’s CZ basis. 114 tests on the core package.

REAL QUANTUM HARDWARE · IBM MARRAKESH · 11 NOVEMBER 2026

On IBM hardware, fewer gates won.

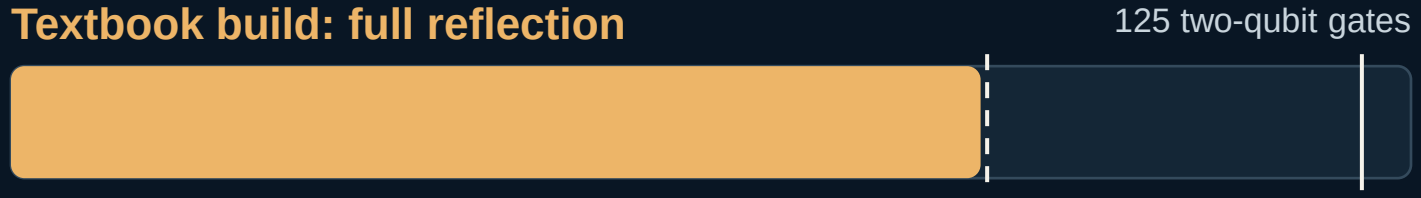
Our lighter build: local reflection



76.90%

6,300 of 8,192 shots hit the click

Textbook build: full reflection



69.26%

5,674 of 8,192 shots hit the click

No amplification: plain sampling



6.74%

solid tick: ideal simulator dashed tick: our noise model (next slide)

+7.6 points

for the lighter build (95% interval 6.3 to 9.0)

0.12 vs 0.18

error in the answer. Lower is better.

7% → 77%

amplification at work on a real chip

8,192 shots per circuit, raw counts, no error mitigation. Job dbqb1ifv5hds73dfo310. Width and routing also differ, so gate count is not proven to be the only cause.

SIMULATOR AGAINST HARDWARE

A simple noise model predicts the gap.

The same compiled circuits that ran on the chip, simulated offline with the chip’s own error rates.

CHANCE OF THE CLICK	Local · 58	Full · 125
Ideal simulator no noise: the two builds tie	96.1%	96.5%
Noise model, calibration only the chip’s reported error rates	83.2%	76.3%
Noise model, echo-fitted gate errors × 1.54, fitted on echo circuits only	78.4%	69.7%
IBM Marrakesh, measured 8,192 shots each	76.9%	69.3%

1.5× noisier

than its calibration said, measured by running the machine forwards and then backwards.

Suppression did not help

Dynamical decoupling and twirling moved the answer error from 0.11 to 0.15 (local) and 0.20 to 0.24 (full).

The bias is predicted

After-click answer, local build: 8.8% and 11.7% measured, 8.4% and 11.3% predicted. Exact: 4.7% and 10.1%.

Depolarizing gate errors from the saved calibration, readout errors measured in the same job. Not modelled: idle decoherence, crosstalk, drift.
hardware_campaign/noise_model/

SEE IT RUN

The Lab: our model, live in a browser.



TRY IT

Scan for the code, the Lab and every result.

- Live, exact simulation
- Real IBM counts
- 25 tests hold it to Qiskit and to the saved data

A 45-second screen recording of lab/index.html. Open Rewind_Amplify_Lab.html to drive it yourself: one file, no install, works offline.

WHO COULD USE THIS

Three places hidden memory matters. Honestly scored.

MODEL RUN ON IBM HARDWARE

Quantum hardware teams

A physical model of the click record lets you compare explanations for a drifting qubit instead of guessing, and tell a defect from a detector error.

0.069 → 0.706

fitted coupling once detector dead time was modelled. True value 0.65 (synthetic records).

SIMULATED · PUBLISHED DATA REANALYSED

Photon-source engineers

Photonic chips need photons that match, not just photons that arrive. Counting clicks misses that.

20% to 100%

photon overlap across 4 model sources with identical brightness.

SIMULATED · CONVENTIONAL SENSOR AHEAD

Laser and lithography metrology

Is the light source drifting, or the sensor? We added an atomic reference. It helped a little. A second ordinary reference helped far more.

58% vs 98%

timely detection: atomic channel against a second conventional reference.

The second and third cards are simulations and a reanalysis of published data. No photons, lasers or wafers were measured by our team.

WHAT WE CAN DEFEND

What we claim, and what we don’t.

WE CLAIM

- A recurrent quantum model made reversible, with identical click statistics.
- Amplitude amplification on its histories, with the answer preserved in the ideal case.
- A lighter, noise-aware build that beat the textbook build on real IBM hardware.
- Every number traceable to saved counts, compiled circuits and tests.

WE DO NOT CLAIM

- A quantum speedup. At this size an exact classical sampler is faster.
- A detected defect. IBM qubits played the roles of a model we programmed.
- Any measurement of photons, lasers or wafers.
- That amplitude amplification is ours. It is established prior work.

NEXT EXPERIMENT

Fit candidate models to held-out, calibrated device records and compare with a strong classical hidden-Markov baseline. Then find where a larger hidden memory makes the classical sampler the bottleneck.

TEAM QUBITS · FLORIDA GULF COAST UNIVERSITY

What we built, and what we learned.

Nicholas Taylor
QRSM research, the Grover fusion, experiments

Timofey Yudintsev
Research, and the keyword-search idea

Dan Izquierdo
Experimentation, and the 45-second video

Dillon Satria Sitepu
Formatting and revising

WHAT WE LEARNED

More hits is not more truth.
A noisy circuit can return the right bit for the wrong reason. Check the answer, not only the hit rate.

On today’s hardware, lighter wins.
58 gates beat 125, in the run and in the noise model.

Report what goes against you.
A conventional sensor beat our quantum add-on, 98% to 58%. It is in the deck.

Code, Lab and evidence

Thank you. Questions?

The QRSM idea and its exact filter predate the event. Amplitude amplification is prior work. Built with AI assistance (Claude, Codex); the team set the questions and checked the results.

Hardware repeats and noise controls

Current run	Shots each	Local raw	Full raw	Conditional TV*
First plain	2048	77.39%	64.26%	0.107 / 0.201
Suppressed	2048	76.46%	61.96%	0.145 / 0.244
Precision follow-up	8192	76.90%	69.26%	0.118 / 0.181

58 / 125 native CZ gates, local / full, in every current run

The follow-up confirms higher yield, with a different effect size

*Raw conditional-history TV, local / full. Lower is closer. Same initial map prefix; widths, routing and times differ.

Deep optical queries show large distribution errors

Three retained optical bins, all-click query, one amplification round on IBM

Choi cutoff	Qubits	Native CZ	Raw accepted	Data Z error*
0.03	15	26,749	0 / 512	0.437
0.003	18	54,998	0 / 512	0.384

Ideal acceptance is only about 0.128%

Zero hits alone is inconclusive at 512 shots. Distribution bias is substantial.

*TV of measured data-register Z probabilities versus the declared circuit target. No optical sensor or full-state fidelity.

Sensor assumptions and resource accounting

Model element	Assumed setting
Compatible laboratory wavelength	698 nm, narrow-line optical drive
Atomic ensemble	64 independent, unentangled atoms
Interrogation	2.5 ms pulse + 20 ms dark time + 2.5 ms pulse
Record and acquisition gap	128 s record, 975 ms dead time per cycle
Relaxation and readout	T_2 = 200 ms, T_1 = 100 s; 2% / 4% readout errors

Time and task match. Total apparatus resources do not.

Photon allocation is assumed bookkeeping. Reference lasers, cooling, trapping and readout costs remain uncounted.

Calibration failures can reverse the conclusion

Scenario	C detection / false alarm	C + Q	C + C
Unseen nominal sessions	51.4% / 5.4%	58.0% / 5.2%	98.0% / 4.1%
Shorter atomic coherence	52.8% / 6.0%	47.3% / 6.1%	98.1% / 5.0%
Primary-reference step	44.0% / 91.5%	45.6% / 84.5%	97.7% / 4.8%
Shared-path step	45.6% / 91.1%	48.2% / 94.1%	56.1% / 100.0%

Shared-path drift is indistinguishable from source drift here

Each cell: detection within 32 s / false alarms per 128 s record. 1,920 null and source records per scenario.

Detector dead time can distort inferred coupling

Fresh synthetic records. True coupling = 0.65. Same data for both fits.

Physical model	Fitted coupling	Held-out score relative to HMM*
Dead time ignored	0.069	−0.4506 nats / record
Blind-bin dynamics included	0.706	+0.0078 nats / record

Modeling dead time improves prediction by 0.4584 nats / record

Corrected parameters still show bias and a dephasing boundary

*24-bin records. Corrected model versus HMM interval spans zero. Synthetic inference, not physical defect diagnosis.

Where the quantum component enters

Layer	What it does	Evidence today
Sensor dynamics	Atomic response to modeled laser drive	Bloch model + digital IBM channel
Diagnostic inference	Combines acquisition records	Classical synthetic benchmark
Rare-history query	Amplifies a condition in a coherent model	Qiskit + actual IBM circuits

A sensing benefit does not establish a computing speedup

Individual channels ran digitally. The full acquisition and diagnostic benchmark remains a separate simulation.

The transfer to lithography needs its own test

Signal	Physical role	Current model
193 nm excimer light	DUV exposure source	Not represented
CO_2 drive laser	Creates a tin plasma for EUV	Not represented
13.5 nm EUV	Plasma emission used for exposure	Not represented
698 nm laboratory laser	Compatible narrow-line atomic probe	Synthetic surrogate

Conventional source diagnostics already exist

Proposed value: test complementary information on synchronized observations, then assess a specific source.

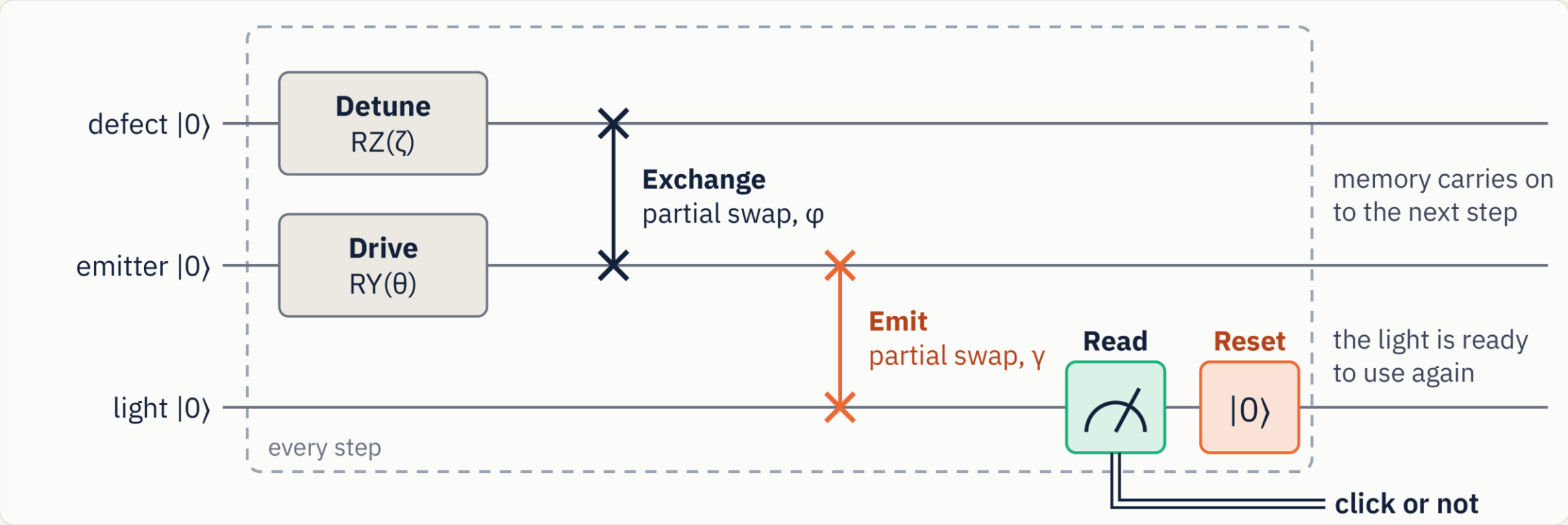
The continued query separates models with bias

Circuit	Defect: step 6*	No defect: step 6*	Difference, points
Plain	13.41%	10.06%	3.35
Steer + local	8.79%	11.66%	-2.87
Steer + full	12.90%	10.87%	2.03
Memory-only control	8.46%	3.97%	4.49

Exact: 4.67% / 10.10%. Local follow-up difference: −2.87 points.

*Raw rate given step-3 click. 8,192 shots before conditioning. Local difference 95% shot interval: −3.89 to −1.84 points.

One step of the recurrent quantum model



Dynamic version: measure and reset the output

Purified version retains a fresh output for each step and defers measurement, preserving the click statistics.

Judging rubric and evidence map

Criterion	Weight	Where the talk addresses it
Problem, and who it affects	15	2, 4, 11: the hidden defect, who has the problem, two futures
Why quantum, and honest limits	20	5–7, 9, 12: cost of rarity, two catches, no speedup today
Technical Qiskit implementation	20	3, 7–8, 10, 22, 38: model, four moves, compiled circuits, Lab
Results and validation	10	4, 8–9, 14, 21, 36: exact, simulator, hardware, noise model
Originality	10	7, 13, 24: reversible recurrent model, noise-aware build
Communication	15	13 main slides, planned 9:20 including a 45-second demo
Q&A, teamwork, learning	10	13–38: roles, lessons, evidence backups

Bonus: simulator +1 (slides 7, 9), real IBM hardware +3 (8), hardware with noise analysis or suppression +5 (9, 14, 36). Evidence supports evaluation, not a guaranteed score.

Contribution, prior work and AI assistance

Origin	What belongs there
Established prior work	Amplitude amplification and amplified conditional inference
Before the event	QRSM idea, physics-shaped model and exact filter
Original hackathon work	Qiskit realization, Grover fusion, steering, local reflection, experiments
Finals extension	Independent model audits, joint-photon model and fresh hardware campaign
AI assistance	Original: Claude. Finals: Codex with OpenAI models

A sensing benefit and a computing speedup are separate claims

Atomic laser-noise spectroscopy also predates this project. No new atomic sensing principle is claimed.

Drive amplitude: a useful negative result

Same 96-bin records. Unknown +20% field step. Detection within 32 bins.

Detector	Timely detection	Stable-record false alarms
Physical QRSM	3.71%	6.84%
Classical HMM	3.13%	5.27%
Count baseline	1.76%	1.95%

Prior-only RMSE 0.1136 → observed-data RMSE 0.1125

Most of the apparent estimation gain came from the prior

Post-holdout prior control is descriptive. 512 records per case; 6,656 total. No physical time or photon-budget calibration.

Phase inspection: benefit depends on the budget

Two illuminated sample modes. All incident photons are charged before loss.

Protocol at 81% efficiency	Full 2π phase RMSE	Wrong π branch
Coherent reference	0.00791 rad	0 / 2,000 trials
Pure two-photon NOON	2.22674 rad	50.4%
10% coherent + 90% NOON	0.00637 rad	0 / 2,000 trials

20,000 mean incident photons; ideal preparation in this comparison

1% preparation: matched dose takes 100× longer for pure NOON

Synthetic phase task, known reference and loss. Hybrid pilots are charged. Zero observed branch errors is not a guarantee.

The full spatial query is costly to construct

49 exposure cells, capped photon counts and a reversible weighted threshold

Constructed bound	Qubits	Generic CX upper bound	Dose-1.5 event probability
Floor weights	389	3,382,108	0.00423713
Ceiling weights	389	2,874,028	0.00422695

196 count qubits + 23 score qubits + 170 extra clean ancillas

Both circuit files round-trip. Classical evaluation is already tractable.

389 qubits exceed the available 156-qubit device. Unoptimized bounds; no full-state simulation or physical optical claim.

Exposure: contrast trades against dose and loss

Phenomenological two-photon resist; fixed threshold and 20 evaluated guard cells

Illumination	Best mean cell accuracy	Incident photons	All 20 cells correct
Two-photon NOON	95.16%	14.88 million	36.76%
Phase-cycled coherent	78.28%	6.81 million	0.73%
10% duty coherent	83.29%	0.845 million	2.52%

Quantum contrast helps this chosen response; it is not dose advantage

Loss, blur and linear background can erase the pattern benefit

Each row uses its own best tested dose. All phase-cycle exposures charged. Higher classical peak intensity is allowed.

Model accuracy is part of the resource budget

Cavity truncation changes conditional predictions, even with exact circuit execution

Cavity levels	Default drive error*	Higher-drive error*
2	2.7643 percentage points	7.0829 percentage points
4	0.0494 percentage points	0.1118 percentage points
5	0.00281 percentage points	0.01102 percentage points

Controlled records improve some model comparisons; ambiguity remains

Classical filtering remains the baseline. More physics costs more qubits.

*Largest tested after-click probability error versus cavity level 6. Finite truncation study, not universal convergence.

Equal photon counts can hide poor interference

Same brightness and one-photon frequency distribution. Different pair correlations.

Frequency relation in a pair	Each photon’s offset	Overlap M	Coincidence per attempt
Shared offset	50% at either offset	1.00	0.0%
Independent offsets	50% at either offset	0.60	0.8%

A joint observation model now tests this broader memory question

80% versus 20% survival: 16× as many attempts per pair at lower efficiency

Attempt = two excitations. Classical environment, ideal preparation, 20% survival. M = 1 is ideal overlap. No quantum-memory advantage.

External GaAs data: a conditional reanalysis

Münzberg et al. (2022) measurements, reused under CC BY 4.0

Excitation separation	Our corrected HOM visibility	Conditional 95% interval
13.1 ns	0.8024	0.7943 to 0.8113
26.2 ns	0.8258	0.8181 to 0.8344
39.4 ns	0.7724	0.7647 to 0.7808

39.4 ns: co-only normalization gives 0.7938

Source diagnosis needs calibrated controls

Our ±2 ns windows and four reference peaks. Published corrections. Conditional counting/calibration errors exclude systematics.

The joint-photon QRSM observation model

A new complete-emission surrogate carries the hidden environment between pairs

Physical step	Digital representation	Retained information
Emit each photon	Complex wavepacket Gram modes	Photon modes + environment
Observe a photon pair	Controlled-SWAP parity = HOM statistic	Conditional environment state
Predict the next pair	Repeated completely positive maps	Dependence on the full record

Simulation, 64 pairs: 5.60% coherent flags, 4.50% classical false flags

IBM four-pair flag-distribution TV: 0.249 coherent / 0.286 classical

TV is raw measured versus ideal flag probabilities, lower is closer. These tests do not identify a physical defect.

More data helps only under trusted calibration

Selected control and ideal resets between independent four-pair blocks

Attempted pairs / record	Nominal coherent flags	Classical false flags
64	11.70%	4.95%
256	19.70%	4.60%
1,024	49.80%	5.90%
2,584	79.75%	4.85%
4,096	92.40%	4.40%
7,644	99.80%	5.20%

The stronger 8-state HMM predicts as well within uncertainty

At 7,644 pairs: changed collection gives 72% false flags, mode overlap 53.25%, readout 21.05%. Each cell: 2,000 records.

Atomic and photon channels on IBM

Actual gate execution of bounded physical models, with raw readout retained

Atomic detuning	Ideal excited probability	Raw IBM probability	95% shot interval
−5 Hz	20.86%	32.32%	30.33% to 34.38%
0 Hz	50.00%	48.24%	46.08% to 50.41%
+5 Hz	79.14%	72.80%	70.83% to 74.69%

Finite pulses and dissipation. 4 qubits, 22 native CZ gates.

Two-photon NOON at phase 0.35: raw occupation TV = 0.033

2,048 shots each. These are superconducting encodings, not measured atoms, photons or a 128-second optical monitor.

Hardware coverage and circuit limits

10 IBM jobs, 207 science executions, 244 charged QPU seconds

Research family	What ran or was constructed	Boundary
Original QRSM	Full original suite, precision follow-up	Bias and routing still matter
Dot / atomic / photon models	Prepared states, channels and joint readout	Digital emulation, no optical setup
Spatial and long queries	Small models plus deeper stress tests	High acceptance can hide wrong answers
Full 49-cell process	389-qubit constructive circuit	Exceeds the 156-qubit device

All 25 research families have an explicit evidence and gap ledger

Classical fitting, published optical data and industrial process calibration remain distinct from superconducting execution.

BACKUP · SIMULATOR AGAINST HARDWARE

The same noise model, circuit by circuit.

After-click answer at step 6. Exact: 4.67% with the defect and 10.10% without. The model was fitted only on echo circuits.

BUILD	TWO-QUBIT GATES	PREDICTED	MEASURED	ORDER ON THE CHIP
Plain sampling	88 / 15	12.5% / 12.1%	**13.4% / 10.1%**	**reversed**
Steer + local reflection	98 / 32	8.4% / 11.3%	**8.8% / 11.7%**	**right**
Steer + full reflection	182 / 56	10.4% / 11.4%	**12.9% / 10.9%**	**reversed**
Memory-only control	93 / 23	8.4% / 3.3%	**8.5% / 4.0%**	**wrong by design**

Each pair is: with the defect / emitter only. Plain sampling fails because false clicks swamp a 4% event.

hardware_campaign/noise_model/results.json. One scale factor stands in for everything the model leaves out.

BACKUP · OPTIONAL COLD OPEN

The cinematic cut. 45 seconds.



A cinematic explainer with original music, not a screen recording of the product. The model picture is conceptual; the IBM numbers are the measured ones.

Anyone can rerun it. One command.

```
$ python reproduce.py

1. The exact model                          with the defect   emitter only
   click at step 6, given one at step 3              4.67%        10.10%

2. The Qiskit circuits on Aer                hit rate   answer error
   plain sampling                               3.95%         0.0000
   steer + full reflection, 1 round            96.49%         0.0000
   steer + local reflection, 1 round           96.10%         0.0353

3. Classical baseline: exact sampler, a fraction of a millisecond per sample

4. IBM Marrakesh, job dbqb1ifv5hds73dfo310      ideal   noise model   measured
   steer + local reflection    58 gates    96.10%     78.40%     76.90%
   steer + full reflection    125 gates    96.49%     69.72%     69.26%

Done in about 10 s. Nothing above claims a quantum speedup.
```

No IBM account, no network. Model numbers are recomputed with Qiskit; measured numbers are recounted from the saved raw counts.